What Didn't Divide

Base as divisor, the digit as residue, and the adder's carry as a limit of pattern selection

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Abstract

Three earlier papers built a reader-side theory of the ripple adder: how many Myhill–Nerode states a reader must hold after seeing part of the input, and when that count factorizes. This paper closes the one step they left open and then follows the carry out of the adder.

The open step closes as a two-block statement. A single unread gap of depth g with $b^g \geq m$ splits the reader's state count into the product of the two sides (455 of 455 instances), and the conditional product law follows by induction. More strongly, the count always factorizes over the blocks that such gaps separate, whatever the narrower gaps do (1,349 of 1,349 cuts). The threshold is shown to mean that the gap can impose *every* carry value, not merely one, and the interface is shown to obey an exact budget, transmit + impose = m + 1. Closed forms are given for the reach and for the entrance ramp of the reading profile.

The carry chain itself turns out to be classical. It is Holte's matrix, and its base-independent eigenvectors and b^{-j} eigenvalues are what the series had been rederiving. What is new is the observer.

A change of base is then read as a whole divided down: the digits are the residues of repeated division and the carry is the quotient in transit. Residues of this kind carry behaviour wherever a repeating unit exists, and the paper asks where the repeating unit comes from. Five mechanisms are given with their mathematics: closure, balance, instability, coupling and history. In a pattern-forming system on a closed domain the preferred period is selected by a growth rate, the allowed periods are fixed by closure, and the mismatch is discharged by inserting or removing one cell, which is a carry. Simulation shows the carry, and shows a coexistence window of width 0.22 in the control parameter within which two cell counts are both stable and history decides between them. The adder has bistability as well, but only at isolated inputs. It is the limit of pattern selection in which the period is handed in, the residue is a whole number, and the coexistence window has shrunk to points.

Every theorem in the series is proved in one dimension, where a boundary cannot talk to itself. That condition is stated here as the governing hypothesis rather than as a caveat. A ledger of thirty-five

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corrections, against the earlier papers, against drafts of this one, and against readings of it, is recorded as part of the result.

Keywords: Myhill–Nerode congruence · carry · positional notation · periodicity · pattern formation · Swift–Hohenberg · Eckhaus instability · hysteresis · Holte matrix · Euler characteristic

1. What this paper claims and what it withdraws

The claims fall into five kinds, and each result below carries one of these labels in substance if not in word.

Proved, or verified by execution. The splitting lemma and the block law; the setter, pointer and separation facts about a gap; the interface budget; closed forms for the reach and the ramp; the quotient induced by top truncation; conservation of carry wires under regrouping; aperiodicity of the carry-transfer monoid; the carry and its hysteresis in a Swift–Hohenberg ring; the arithmetic of converting π's hexadecimal expansion to decimal. Every count quoted is from an executed run and is listed in §16.

Identified with existing mathematics and credited. The carry chain is Holte's matrix [5], with the shuffle and representation theory of Diaconis and Fulman [6, 7]. The building blocks of the periodic table are spherical harmonics. The twelve pentagons of a closed hexagonal mesh are Euler's formula. The forced halts of a flow on a sphere are Poincaré–Hopf. The discharge of mismatch in a periodic pattern is the Eckhaus phase slip [14]. None of these is claimed.

Named as a hypothesis. Every theorem in the series is proved where the boundary of a region is two isolated points and an interface has one parameter, its length. §10 states this as the governing condition of the corpus.

Offered as interpretation, and labelled as such. That the carry is the narrow-window limit of pattern selection; that a residue carries behaviour wherever a repeating unit exists; that the observer theory of the adder can be read as an observer theory for the operator underlying Holte's matrix.

Withdrawn. The reading of the adder as a model of a persistent thing; the language of loss that runs through the earlier papers; and a number of claims made in drafts of this work, each logged in §13.

The relation to the earlier work is direct. *Two Quotients of One Machine* [1] and *What a Cut Must Do to Make a Thing* [2] established the reader count F(S) and the cut law for the two-operand binary adder. *Carry Capacity and the Conditional Cut Law* [3] generalized the machine to m operands in base b and found the product law conditional on the gaps. An unpublished working draft, *The Inversion and Its Empty Cells*, is superseded here; its five-slot specification is kept in §12 and its algebraic sections are reduced to §11.

2. The machine, the reader, and one division

2.1 The machine family

The machine is the m-operand base-b ripple adder. Positions run q = 0 … n−1, the digit sum at each position is $t_q \in \{0, \ldots, m(b-1)\}$, and there is one loop:

$$s_q = (t_q + c_q) \bmod b, \quad c_{q+1} = \lfloor (t_q + c_q) / b \rfloor$$

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The carry takes exactly m values, {0, …, m−1}, at every radix; that is Proposition 1 of [3]. The line machine pins $c_0 = 0$ and discards the carry out of the top position. The ring machine identifies c_n with c_0. A reader is a cut S of the positions and sees t_q for $q \in S$ only. F(S) is the number of Myhill–Nerode classes of what it has read: a state is the set of pending output positions together with their dependence on the unread data, and digits already emitted are gone.

Two results of [2] recur below and are distinguished from [3]'s Proposition 1 to avoid a confusion that crept into drafts of this work. Proposition 1 of [2] says F(S) = 1 exactly when S is empty or the whole word; [3] showed it holds only with the origin pinned, since with c_0 free the full cut costs (b−1)n + 1. Corollary 4 of [2] says least-significant-first is the unique reading order minimizing both the peak state count and $\Sigma \log_2 F$.

2.2 One division read from every side

Any unread run of g positions acts on the carry passing through it. By the nested-floor identity $\lfloor \lfloor x/a \rfloor / d \rfloor = \lfloor x/(ad) \rfloor$ the action always has the same form, Lemma 1 of [3]:

$$\tau_{g,A}(c) = \lfloor (c + A) \,/\, b^g \rfloor, \quad c \in \{0, \ldots, m-1\}, \;\; 0 \le A \le m(b^g - 1)$$

Here A is the base-b value of the gap's word. Composition closes on this form: a depth-q transfer after a depth-g transfer is a depth-(q+g) transfer. The entire quantitative content of the series is this one division, and six results that were found separately, in different corners, are readings of it.

Reading	Quantity	Where it was found
Interface transfer	$\tau(c) = \lfloor (c+A)/b^g \rfloor$	[3] Lemma 1
Severance condition	$b^g \ge m$	[3] §4, Lemma 2
Surviving distinctions	$\lceil m/b^g \rceil$	minimum image size at depth g
Imposable outcomes	$m - \lceil m/b^g \rceil + 1$	reach from the pinned origin
Remainder capacity	$\min(m, b^k)$	[3] §5.1, the tail cap
Spectral contraction	eigenvalues b^{-j}	Holte [5]; §6 below

The fifth reading is the other half of the division. A gap keeps the quotient $\lfloor (c+A)/b^k \rfloor$; a top truncation with k positions remaining keeps the remainder. Running it confirms that two carries are indistinguishable by the remaining positions exactly when they agree modulo b^k (722 of 722 cases). The two ranks multiply back to the alphabet: $\lceil m/b^k \rceil \cdot \min(m, b^k) \ge m$, with equality exactly when b^k divides m.

This is why every threshold in the series is the same number. The entrance ramp of the least-significant-first profile, the gap depth needed to sever two runs, the depth at which the reachable set saturates, and the depth at which a constant transfer first appears are four events, and each occurs when $\lfloor (m-1)/b^g \rfloor$ reaches zero.

2.3 Closed forms for the reach and the ramp

The number of carry values a depth-L window can build from the pinned origin is

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$$\mathrm{reach}(b, m, L) = m - \lfloor (m-1) \,/\, b^{L} \rfloor$$

The proof is induction on the nested-floor identity: with $q = \lfloor (m-1)/b^{L} \rfloor$ the next worst case is $\lfloor (m(b-1) + m - 1 - q)/b \rfloor = m - 1 - \lfloor q/b \rfloor$, and $\lfloor q/b \rfloor = \lfloor (m-1)/b^{L+1} \rfloor$. It holds in 1,848 of 1,848 cases at $b = 2 \ldots 12$, $m = 1 \ldots 24$, $L = 0 \ldots 6$. The reach saturates at m exactly when $L = \lceil \log_b m \rceil$, in 418 of 418 (b, m) pairs, which is the gap threshold arriving on a different object. Together with the tail cap this gives the reading profile in closed form:

$$F_L = \min\left(m - \lfloor (m-1)/b^{L} \rfloor \,,\ \min(m, b^{n-L}) \right)$$

It reproduces the machine exactly on every configuration tested, including the profile 1, 3, 4, 5, 4, 2, 1 at b = 2, m = 5, n = 6 that [3] reported without explanation, and the integrated costs 5.0000, 6.7549, 8.5850, 8.9069 and 4.7549. The ramp is the reach, the descent is the tail cap, the plateau is m. This closes open problem 2 of [3].

At m = 2 and b = 2 every one of these readings collapses to a constant. The threshold is 1, which every admissible cut satisfies; the ramp is one step; the tail cap and the alphabet are both 2. Six laws reduce to four constants and a length, and nothing in the data suggests they are related. That is the standing pattern of the programme, and §15 returns to it.

3. The composition step, closed

3.1 The splitting lemma

[3] left one step open: that separation at each gap composes into the global product over all runs at once. It closes here, and it closes as a statement about two blocks rather than k runs.

Splitting lemma. Let $S = S_{lo} \sqcup S_{hi}$ with $S_{lo} \subseteq [0, p-1]$ and $S_{hi} \subseteq [p+g, n-1]$, both non-empty, and positions $p \ldots p+g-1$ all unread, with $b^{g} \geq m$. Then $F(S) = F(S_{lo}) \cdot F(S_{hi})$.

It holds in 455 of 455 instances across 27 (b, m, n) configurations up to n = 8. The product law of [3] follows by induction: split at the lowest qualifying gap and recurse on the upper part, with a single run as the base case. This closes open problem 1 of [3]. The proof rests on three facts about the interface, of which only the first was in [3].

The upper bound is free. The residual function of a read assignment factors through the pair of block states, so $F(S) \leq F(S_{lo}) \cdot F(S_{hi})$ at every gap, wide or narrow; 1,151 of 1,151 splits, no violation. Across the whole sweep the run-product overcounts 431 times and never undercounts. Only the lower bound ever needed a threshold.

The gap can set. If $b^{g} \geq m$, then for every $v \in C$ the word $A = v \cdot b^{g}$ realises the constant map $c \mapsto v + \lfloor c/b^{g} \rfloor = v$, and A is representable because $(m-1)b^{g} \leq m(b^{g}-1)$ exactly when $m \leq b^{g}$. Both conditions bind at the same place. Across 105 (b, m, g) triples the set of carries a gap can force is either all of C or empty, never in between. So the threshold does not mean that the gap can realise a constant. It means the gap can realise every constant. [3] displays this in its own table, where the realisable constants are listed as [0, 1, 2] and [0, 1, 2, 3, 4], and reads it as the existence of one.

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The gap can point. The word $A = b^g - 1 - c$ realises the threshold map $x \mapsto [x > c]$, which is available whenever $b^g \geq m - 1$ (105 of 105). Any two distinct carries leaving the lower block can therefore be mapped to 0 and 1, which differ modulo b, so the difference shows in the output at position p+g whatever is read above it.

With these, distinct upper-block states are separated because the gap can reproduce any carry the standalone cut used, and distinct lower-block states are separated either below p+g, where nothing above matters, or through the pointer. Hence $F(S) \geq F(S_{lo}) \cdot F(S_{hi})$, and with the upper bound the lemma follows.

One further fact belongs here because it does not follow the threshold. At every depth, including depth one in every base, some gap word distinguishes any chosen pair of carries; 105 of 105 triples. The threshold controls whether an interface can make two states the same, never whether it can tell them apart.

3.2 The block law

The lemma is stronger than the conjecture it settles, because it is indifferent to the other gaps.

Block law. $F(S) = \prod F(B)$ over the maximal blocks of S separated by gaps of depth at least $\lceil \log_b m \rceil$, for every cut, whatever the narrower gaps do.

It holds for 1,349 of 1,349 cuts across the same configurations: 919 single-block, 405 two-block and 25 three-block. When every gap is wide, every block is one run and this is [3]'s conditional product law. When some gaps are narrow the cut still factorizes into the coarsest pieces the wide gaps define, and the residual coupling is confined inside a block. This locates the one-way implication recorded as correction 12 of [3], in which 84 cuts satisfied the run-product with no constant available anywhere and none failed with a constant available everywhere: a cut can satisfy the run-product inside a block by accident, without any gap clearing the threshold. §6.2 gives the quantitative reason the accident is common.

3.3 What remains argued rather than proved

The upper bound is argued through the bookkeeping of pending and emitted digits rather than written as an induction on n, and that bookkeeping has already produced one false statement. A draft identified the lower block's reader state with its interface, meaning the varying outputs below p together with the carry at p. Running it: false, 96 of 111 agree. The interface is strictly finer than the state, because carries congruent modulo b^{n-p} cannot be told apart by what remains above. Repaired with that quotient the identification is exact, 148 of 148. The lemma does not rest on the false version, but the episode is why the upper bound is labelled argued. Writing it as an induction is open problem 6.

4. The interface budget

4.1 Transmit and impose

A gap does two things to the carry that crosses it. Write transmit(g) for the number of incoming carry distinctions that survive a depth-g gap, and impose(g) for the number of outgoing carries the gap can produce from a fixed incoming one by its own choice of word. Then

$$\text{transmit}(g) = \lceil m / b^g \rceil, \quad \text{impose}(g) = m - \lceil m / b^g \rceil + 1, \quad \text{transmit}(g) + \text{impose}(g) = m + 1$$

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Both are the single quantity $\lfloor (m-1)/b^g \rfloor$ entered with opposite sign, and impose(g) is the reach of §2.3 under a second reading: what a g-word can build from zero is exactly what it can force onto a downstream carry, since in both cases the word supplies the offset A and the depth supplies the divisor. At g = 0 the gap is pure conduit, transmitting m and imposing 1. At $g = \lceil \log_b m \rceil$ it is pure source, transmitting 1 and imposing m. The threshold is not where the carry's information dies. It is where the budget finishes crossing over.

4.2 Separation is purchased

[3] says throughout that a wide gap forgets the carry. Under the identity that is half a transaction. At a gap wide enough to sever two runs, the m distinctions that stopped flowing through the carry are held by the gap word — exactly m of them, since the forced set is all of C or nothing. The field between the runs is where the crossing state went.

The same correction applies to the resolution tower of [3] §8, which reports coarsening as a fibration with constant fibre $2^{N-N/k}$ and calls the uniformity an evenly spread loss. The reverse is the case. Since log|fine| = log|coarse| + log|fibre| exactly, a uniform fibre is a change of units. Loss would require a fibre that varies with the coarse state, some coarse states hiding more than others, and [3] measured that it does not. §6.1 gives the structural reason it cannot.

4.3 Selection cannot buy depth

A draft of this work stated that pointing is free while forgetting costs $\lceil \log_b m \rceil$. That was accounting with the boundary drawn where it flattered the result. Charge for the gap word and both directions cost: naming a word at depth g takes about g ln b + ln m of log-cardinality, and the word that forgets the whole alphabet costs more to name than the ln m it removes.

What survives is a difference in kind rather than in price. The separating word is already present at depth one and needs only to be selected. No selection of A produces a constant below $\lceil \log_b m \rceil$. Selection cannot substitute for depth; that is a structural bound, not a cost.

The author's bucket makes the point without arithmetic. Gravity fills and empties a bucket for nothing if the water is already above the rim, and in that sense filling and emptying cost the same. But no way of pouring makes a gallon bucket hold two. Capacity is structure; fill is selection. Forgetting is not expensive — it is unavailable until the structure is there, and then it is as cheap as pointing.

4.4 Two currencies

The conservation in §4.1 is in cardinality and does not survive the logarithm: ln(transmit) + ln(impose) ≠ ln(m+1). A second reading is logarithmic and measures cost rather than allocation. The carry alphabet has log-cardinality ln m. This is Hartley information and not Shannon entropy: no probability measure is assumed anywhere, and §6.2 shows the actual distribution of the carry is not uniform. One gap position divides the carry by b and removes ln b of it, so

$$g_* = \lceil \ln m / \ln b \rceil = \lceil \log_b m \rceil$$

which is how much must be forgotten over how much one position can forget. The two ledgers should not be blended. Cardinality says where states are allocated; log-cardinality says what collapsing them costs; the rate of exchange between them is not linear.

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4.5 An interface is an abstraction boundary

A method can be called without knowledge of its implementation exactly when the interface between them resets, so that nothing from below leaks through. Below the threshold it does not reset, the abstraction leaks, and the caller must know the layer beneath. The severance condition is therefore the law of leaky abstractions with an exact criterion attached, and a price for meeting it (§5.4). This is the precise content of reading the sciences as one large programming interface, and it is also its limit: the unity holds where interfaces sever and has to be checked for each pair of domains, not assumed.

5. A change of base is a whole divided down

5.1 Digits are residues

A number does not change when its base does. What changes is how it is divided. Divide the whole by the base and two things come out: a remainder, which is what the base could not absorb, and a quotient, which is everything else and is passed forward. Divide the quotient again, and again, until nothing is left. The digits are the remainders. The quotient is the state in transit, starting as the entire value and draining to zero, and it is the carry register: base conversion is the adder with the value loaded in the register, which [3] §7 identified and filed as a curiosity. The number of divisions is the log of the whole over the log of the divisor.

So a representation is not built up out of digits. It is the whole subdivided, and a digit is the part of it the divisor failed to take. Nothing accumulates. Changing the base divides the same whole by a different number and different residues fall out. That numbers are positions in a structure rather than objects in their own right is the structuralist position in the philosophy of mathematics [4]; the division picture is a mechanical version of it.

5.2 The worked case: π from hexadecimal to decimal

The first twenty-two hexadecimal digits of the fractional part of π, read as an integer, are

$$(243\mathrm{F6A8885A308D313198A})_{16} = 4382080378686747050738317 8 = \lfloor \mathrm{frac}(\pi) \cdot 16^{22} \rfloor$$

and the conversion is exact. The mechanism is visible in the two bases' factors: $16 = 2^4$ and $10 = 2 \cdot 5$. They share a 2 and differ by a 5, and the split governs how the transformation behaves.

The shared part is local. The value modulo 2 is the parity of the last hexadecimal digit; A is even, so the value is even, and nothing else contributes.

The unshared part is global, and it has a shape. Because $16 \equiv 1 \pmod 5$, the value modulo 5 is the sum of its hexadecimal digits modulo 5 — casting out fives, the hexadecimal counterpart of casting out nines. The twenty-two digits sum to 138, which is 3 modulo 5. The last decimal digit is therefore fixed by the parity of A and the digit sum of the whole string: the only digit that is even and 3 modulo 5 is 8, and the decimal ends in …178.

The locality difference can be measured directly. Changing the single top hexadecimal digit changes one binary digit and twenty-three of the twenty-six decimal digits. Binary is a regrouping of hexadecimal, four bits to a digit with no carry crossing a group; decimal cannot be reached that way, because its 5 has to be manufactured from every digit at once. In general, primes the two bases share carry across locally, primes

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they do not share force global mixing, and when the old base is 1 modulo the new prime the mixing collapses to a digit sum.

5.3 Regrouping conserves every wire

Regroup n fine positions into n/k blocks of k. Inside each block there are k−1 carry crossings; between blocks there are n/k − 1. The total

$$(n - n/k) + (n/k - 1) = n - 1$$

is the fine machine's own crossing count. Coarsening removes no carry dependency. Each is reclassified from a wire between digits to a wire inside one, and a base-b^k digit is a k-position base-b adder with its carries placed behind the interface. What regrouping asks for is simultaneity: a base-b^k position is k base-b positions held at once.

5.4 What a change of base costs

There are three ways to hold k digits at once, and they are the three axes an implementation has.

Resource	Cost of one base-b^k digit	Scaling in k	Status
Space	k parallel lines	linear	exact
Sequential time	k clock intervals	linear	exact
Settling time	relaxation to 1/L of final value, $L = b^k$	$k\,\tau \ln b$, linear	exact for first-order relaxation
Signal energy	L-level signalling at fixed error rate	about b^{2k}, exponential	assumed model, $SNR \propto L^2$

Time is linear in k on both routes. First-order relaxation reaches accuracy 1/L in $\tau \ln L$, and $\ln L = k \ln b$, so exponential decay converts an exponential count of levels into a linear time. The amplitude route is exponentially expensive only in energy: margin falls like b^{-k} while the digit carries $k \log_2 b$ bits, so energy per bit grows like b^{2k}/k against a flat serial baseline. Small-radix digital hardware is favoured by energy, not by latency. The last two rows are models applied on top of exact combinatorics, and the step from one to the other is not a theorem about the adder.

A spirit level presents the far end of this dial, one position read as a static shape. A draft said its medium supplies resolution for free. It does not: it pays in restoring force, damping, settling time, tolerance, drift and stored energy. What it does is move the price to an axis the physical configuration pays once, rather than per symbol per cycle.

The price of a boundary picks a base. Charge one position in proportion to the states it must resolve, $m(b-1)+1 \approx mb$. A separating gap needs $\lceil \log_b m \rceil$ positions, so

$$\text{cost}(b, m) \propto b \log_b m = (b / \ln b) \cdot \ln m$$

which factors into what must be forgotten, ln m, and what the representation adds, b/ln b, minimized at b = e. Among integers b/ln b is 2.885 at base 2, 2.731 at base 3, and 2.885 at base 4, where the tie is exact because 4/ln 4 = 2/ln 2. The expression is classical radix economy; the route is not. It arrives as the price of

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building an interface that can forget, rather than of storing a number. The model counts state resolution and nothing else, and is not an argument that ternary hardware is cheaper.

Redundancy buys depth one. In a signed-digit representation with digits $\{-a, \ldots, a\}$ and $a > \lceil (b-1)/2 \rceil$, two-operand addition is carry-free: a transfer reaches one position and stops, at every base [9]. The price is an alphabet roughly twice as wide and the loss of unique representation — many digit strings for one value, which is a fibre installed at every position instead of hidden in the tower. That is the strongest evidence that $\lceil \log_b m \rceil$ is a property of non-redundant representation rather than of addition.

Regrouping needs an origin. A cut survives regrouping by k exactly when its boundary lies in $k\mathbb{Z}$, and divisibility by k presupposes a zero. The location invariant loc(S), the gcd of a cut's boundary positions with N, is measured from the pinned end ([3] §8, verified over 25,904 cut-resolution pairs). Physically that zero is a mod-k phase counter holding $\lceil \log_2 k \rceil$ bits, one register per rung. On the ring there is nowhere to preset it: blocking by k is preserved only by rotations through multiples of k, so regrouping breaks $\mathbb{Z}/n$ down to its subgroup of order n/k, and the k lost rotations are the choices of phase inside a block, which is what becomes internal to the digit. The ring's rotation invariance ([3] §5.3, 32 of 32 cuts) and the line's divisor tower cannot both be had.

5.5 What the read head reads

The Bailey–Borwein–Plouffe series [8] extracts hexadecimal digits of π at a chosen position without computing the earlier ones, and run here it reproduces all twenty-two of the digits above that way. The reason it lives in base 16 is the mechanism of §5.2. The series is a sum in powers of 1/16. Multiplying by 16^n and discarding the integer part is a remainder taken in the series' own base — the remainder side of the one division, computed without ever forming the quotient. Base 2 works as well because it is a regrouping of 16. Base 10 requires the unshared 5, and the unshared prime is exactly what makes every digit depend on every other, so the locality the extraction uses does not survive the conversion.

No series of this type is known for π in base 10. The argument above explains why the base-16 series does not transfer; it is not a proof that no base-10 series can exist. The question the programme has carried since its start — a read head where, reading what — has a plain answer: it reads $16^n\,\pi$ modulo 1 in the one base where shifting and discarding is an exact operation on the series itself.

6. The machine is classical; the observer is not

6.1 Holte's matrix

The carry of m-summand base-b addition with uniformly random digits is a Markov chain on m states, and its transition matrix is Holte's amazing matrix [5]. Its known properties are, one for one, results the series derived independently.

Classical fact	Where the series rederived it
The chain is m × m at every base	[3] Proposition 1
Eigenvalues $1, b^{-1}, \ldots, b^{-(m-1)}$	the contraction $\lfloor c/b^g \rfloor$ of [3] Lemma 1

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Classical fact	Where the series rederived it
Eigenvectors independent of b	base-invariance of the transfer family
Stationary law A(m,k)/m!, Eulerian	not found by the series
Same matrix as the descent chain of b-riffle shuffles	not found by the series
Left eigenvectors Foulkes characters, right Eulerian idempotents	not found by the series

The shuffle correspondence and the representation theory are due to Diaconis and Fulman [6, 7]. These should be cited as background. The series' deterministic shadows of them are rediscoveries, recorded in the same spirit as an earlier entry in the programme in which an elastic-ring tour solver rediscovered the self-organizing map: evidence about the reasoning, not a result.

Holte's language corrects a draft error. A draft observed that the transfer family at base b and the family at base b^k are the same set of maps, since $B^{g/k} = b^g$ and the alphabet is base-free, and concluded that refinement exposes nothing and the machine has no depth. The set equality is right and the conclusion is not, because depth is indexed by a filtration and a set equality cannot see the index. In an eigenbasis the statement is exact: **the eigenvectors are the family and do not depend on the base; the eigenvalues are the grading and are b^{−j}.** A change of base fixes the basis and rescales the spectrum. The threshold's own behaviour, $\lceil \log_b m \rceil$ becoming $\lceil (\log_b m)/k \rceil$, is that rescaling measured.

The same fact explains a measurement [3] recorded without accounting for it: that the coarsening fibre is uniform. A non-uniform fibre would mean some coarse states conceal more capability than others, and there is no capability to conceal. What a coarse rung hides is which fine word produced a given transfer — provenance, not power — and provenance is spread evenly because the words are.

Two cautions. The deterministic floor map and the stochastic eigenmode are different objects, an image-and-rank structure against a linear one; the safe statement is that the two carry the same radix grading. And the m-dimensional carry space is not itself the group algebra of the symmetric group; the eigenbasis is realised through Eulerian idempotents, which is a realisation of the decomposition rather than an identification of spaces.

6.2 Worst case against typical

The stationary carry law is Eulerian, independent of b, with mean (m−1)/2 and variance (m+1)/12. The carry therefore concentrates on an interval of width of order √m instead of spreading over its m values, and its Shannon entropy is about ½ ln m against the Hartley ln m. That gives two scales where the series used one: guaranteed severance at $\lceil \log_b m \rceil$, exact and distribution-free, and typical decoupling near $\tfrac{1}{2} \log_b m$, a concentration heuristic.

This is a quantitative account of the one-way implication of [3] correction 12. The threshold is a guarantee against the extreme carries, and the Eulerian law makes those rare, so most run signatures present in a cut sit in the concentrated middle, which a shallower gap already decouples. The existing sweeps are consistent without being decisive: at b = 2, m = 5, threshold 3, the crossover from mostly failing to mostly holding is at gap depth 2; at b = 2, m = 3, threshold 2, it is at 1. The test is the fraction of cuts that factorize plotted against

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$g / \log_b m$ across many (b, m), which should collapse onto one curve with its knee near one half (open problem 4). Bounding this chain's convergence from its eigenstructure is known to be awkward, since the eigenvectors are not orthonormal; the one half is concentration, not a mixing theorem.

6.3 The threshold is also a shuffle

Because the carry chain is the descent chain of b-riffle shuffles on m cards, the reset depth has a second reading: the number of unread positions needed to sever two runs is the number of b-shuffles needed to wash out the descent structure of m cards. Severance and randomisation are one operator read from its deterministic and its probabilistic sides. That is a structural comparison, not an established equivalence between reader factorization and mixing, and it will not become one until the curve of §6.2 has been measured.

6.4 The observer axis

Holte studies the chain. This series studies what a partial observer of the chain must retain, which is a different question and, as far as this paper can determine, is not in the carries literature. The observer's objects are F(S); the cut law and its failure conditions; the splitting lemma and the block law; the transmit–impose budget; the dependence of the count on the pinned or free origin and on line or ring; regrouping as a change of the reader's resolution; and reading order as a memory–time allocation, with least-significant-first uniquely minimal.

The cleanest sign that the two are different is that the stationary law is independent of the base and F is not: it depends on b through the run factors $(b-1)L + 1$ and $2(b-1)L + 1$, through the tail cap, and through the threshold. The framing offered is that the cut law *can be read as* an observer theory for the carry operator underlying Holte's matrix — can be read as, not is, since the identification is this paper's, not the literature's.

7. Residues in the world

7.1 Four fields, one decomposition

Wherever a system has a repeating unit it decomposes as $x = qP + r$: a repetition index q saying which unit, and a local phase r saying where within it. The earlier sections read the arithmetic version. Here are four places the same decomposition does physical work. They are not four metaphors for one another; they are four implementations of periodic decomposition, with limits marked.

Chemistry. For the main-group elements, the period is the principal shell and the group is the occupancy of the outer shell, and the chemistry lives in the second. Lithium, sodium, potassium, rubidium and caesium share a single outer electron ($3 = 2 + 1$, $11 = 2 + 8 + 1$, $19 = 2 + 8 + 8 + 1$) and behave alike across a large range of size. Noble gases have a closed outer shell and are inert. The octet rule is atoms driving their outer occupancy to a closed shell, one by giving electrons and one by taking them. And the decomposition says why chemistry cannot turn one element into another: chemistry changes electronic configuration within a fixed nucleus, whereas transmutation changes the atomic number itself. The limit is the transition metals, where the filling of ns and $(n-1)$d shells interleaves and the simple picture fails; §8.1 gives the reason.

Crystals. Position in a crystal is a lattice-cell index plus a position within the cell, and wavevector is defined only modulo a reciprocal-lattice vector G. When two lattice vibrations collide and their combined crystal

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momentum passes the zone boundary, $k_1 + k_2 = k_3 + G$: the reduced part stays in the vibration and the multiple of G is taken up by the crystal as a whole. This is the Umklapp process, and it is the adder's structure exactly — the residue stays in the excitation and the carry goes to the lattice. It is also why heat conduction in a real crystal is finite: the carry is where the momentum leaves.

Calendars. The tropical year is not a whole number of days, so the remainder never closes and the calendar needs periodic leap-day corrections. That is the hexadecimal-to-decimal situation of §5.2 in time rather than space: the part the two cycles do not share forces a periodic global correction. The calendar works not because the remainder is absent but because it is retained and discharged on a schedule.

Genes. A coding sequence is read in frames of three, so the reading frame is position modulo 3. A frameshift changes the phase and every codon downstream is re-divided. The nucleotides are unchanged; the partition of the same sequence has changed. That is the mod-k phase counter of §5.4 off by one, and it is what §5.2 measured when changing one top digit rewrote twenty-three of twenty-six decimal digits.

A draft summarized this as: behaviour lives in the remainder and scale in the quotient. That is too strong. The correct general statement is only the decomposition itself, repetition index plus local phase, with each field choosing which part it makes observable.

7.2 Flat transport and curved closure

All four cases above live in flat transport. In flat space translations commute, so a direction works as an address, and that is positional notation exactly: a digit means something because its place value b^k does not depend on how the place was reached. The crystal case even lives on a torus, since the Brillouin zone with its opposite faces identified is flat, with Euler characteristic zero. The residue picture of §7.1 holds exactly where transport is trivial.

Where transport is curved the residue does not disappear; it moves from a position to a loop. In the Aharonov–Bohm effect [17] an electron passes around a region it never enters and returns with a phase set by the magnetic flux inside, counted modulo the flux quantum. In Berry's geometric phase [18] a quantum state carried around a closed cycle returns with a phase that depends on the loop's geometry and on no point along it. In the flat case the digit is what didn't divide. In the curved case the phase is what the loop didn't close.

8. Where a period comes from

Base conversion was clean because its divisor is handed in. In physical systems the period is a result. Five mechanisms produce one, and each has its own mathematics.

8.1 Closure quantizes

An electron is not moving on a sphere, but the angular part of its state is a function on one, and single-valuedness on a closed domain admits only the spherical harmonics Y_l^m, with $l = 0, 1, 2, \ldots$ and $m = -l, \ldots, l$ — so $2l + 1$ angular states at each l. Two spin states double this to 2, 6, 10 and 14, the capacities of the s, p, d and f subshells. The discreteness that makes the periodic table periodic is the closure of the angular domain.

Closure supplies the building blocks and interaction sets their order. For hydrogen the levels with $l < n$ are degenerate, which gives shell totals $2n^2 = 2, 8, 18, 32$; that degeneracy comes from a symmetry peculiar to

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the 1/r Coulomb potential. In atoms with many electrons, repulsion and shielding split and reorder the levels, 4s fills before 3d, and the observed period lengths are 2, 8, 8, 18, 18, 32. That reordering is where the transition-metal difficulty of §7.1 comes from.

8.2 Balance selects

No one tells atoms the lattice constant. Among continuously variable separations the interaction energy has a minimum, and the crystal forms at that spacing: a system that is the same everywhere as a liquid becomes the same only in steps. The period is the scale at which attraction and repulsion come out even.

8.3 Instability selects

Perturb a uniform state with a wave of wavenumber k and it grows or decays at a rate σ(k) given by linearizing the dynamics. The period that forms is the wavenumber at which σ is largest. The standard model of pattern formation, the Swift–Hohenberg equation [13], $\partial_t u = ru - (\nabla^2 + k_0^2)^2 u - u^3$, has

$$\sigma(k) = r - (k^2 - k_0^2)^2$$

with its peak exactly at k_0. Turing's reaction–diffusion version [12] obtains σ(k) from the reaction Jacobian and the diffusion matrix, $J - k^2 D$, and stripes or spots appear when a band of nonzero k goes positive. Heat a fluid layer from below and past a critical point it breaks into convection rolls of a definite width. This mechanism exists only while something drives the system, and it creates a period from a gradient with no prior oscillation.

Swift–Hohenberg is a gradient flow, so near onset the fastest-growing mode is also the direction of steepest descent in its energy functional: instability and balance select the same k_0 there. Far from onset, or under non-variational forcing, the fastest-growing mode and the eventual stable pattern need not coincide.

8.4 Coupling locks

Two independent cycles stay incommensurate unless something couples them. For a circle map $\theta_{t+1} = \theta_t + \Omega - (K/2\pi) \sin 2\pi\theta_t$ the rotation number $\rho = \lim (\theta_N - \theta_0)/N$ is the mean phase advance per cycle, and coupling produces intervals of the drive in which ρ is rational — the relation locks and closes after a finite joint cycle [15]. The Moon is locked 1:1, its rotation equal to its orbit. Mercury is locked 3:2, three rotations for every two orbits, so $2T_{orbit} = 3T_{spin}$. Earth is not locked, and its calendar residue never closes.

8.5 History freezes

On a closed loop write a pattern's order parameter as $A(x) = |A(x)|\, e^{i\phi(x)}$. While the amplitude is nowhere zero, the number of cells is a winding number,

$$n = (1/2\pi) \oint \partial_x \varphi \, dx \in \mathbb{Z}$$

and it cannot change continuously. To change it the amplitude must vanish somewhere so the phase becomes undefined, which is a defect, and a defect costs energy. So a count, once established, persists. History is a winding number held in place by the price of changing it. The three-letter codon and base ten are the cultural cases: a period fixed early that everything downstream came to depend on.

Taken together, a period is a recurrence that has acquired a mechanism for remaining a recurrence — topological closure, an energy minimum, a growth-rate maximum, a dynamical lock, or an inherited constraint.

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9. The carry as a limit of pattern selection

9.1 A preferred count against an allowed branch

Put §8.1 and §8.3 together on a closed domain. Instability or balance selects a preferred wavelength $\lambda_* = 2\pi/k_0$. On a loop of length L closure admits only $k = 2\pi n/L$. So

$$L / \lambda_* = n + \delta$$

where n, the number of cells, is an integer and δ is the mismatch. This resembles N = qB + r and a draft wrote it as that identity. It is not the same type. In base conversion both q and r are integers and $0 \le r < B$. In the pattern the integer is the branch n, and the mismatch δ is continuous strain. The discrete object is the branch, not the residue.

9.2 The carry appears

The Swift–Hohenberg equation was run on a periodic domain of length 16π at r = 0.3, with k_0 ramped slowly from 1.0 to 1.5, so that the preferred cell count L/λ_* slides continuously from 8 to 12. The realized count never takes a fractional value. It steps 8, 9, 10, 11 in whole units, at $k_0 = 1.185$, 1.294 and 1.408. Between steps the mismatch drifts out to between about 1.0 and 1.2 cells and snaps back when a cell is inserted. A near-onset estimate of the Eckhaus limit [14], the known bound on how far a stripe pattern can be stretched before it breaks, gives about 1.1 cells at these settings. These figures are specific to this model and these parameters; there is no universal slipping threshold.

The mismatch accumulates, a stability boundary is crossed, one unit of the count changes, and the mismatch is reduced. That is a carry.

9.3 The pattern remembers the way it came

Ramping k_0 back down from 1.5 to 1.0 gives different switching points.

Transition	k_0 on the way up	k_0 on the way down	Coexistence window
9 ↔ 10 cells	1.294	1.071	0.223
10 ↔ 11 cells	1.408	1.212	0.196

Between the two switching points of a pair both counts are stable, and which one is observed depends on the direction of approach. At $k_0 = 1.0$ the ring held 8 cells on the way in and 9 on the way out: the same control setting, a different state, and the only difference is the path. The width $W = k_0^{\uparrow} - k_0^{\downarrow}$ is a direct measure of retained branch memory. It is a range of the control parameter, not a duration — a draft read it as how long a count is kept — and converting it into a retention time requires a sweep rate and a noise level.

9.4 The adder has memory too

A draft claimed the adder has no dependence on history. That is wrong: the carry is the adder's history. At b = 2 with two operands, the digit sum 1 gives the carry map $c \mapsto \lfloor (1 + c)/2 \rfloor$, which is the identity on {0, 1}. Held at that input, whichever carry the machine holds, it keeps. Same input, two stable states, and the path decides. It is the same fixed-point structure that gives the ring its residual of 2 ([3] §5.3; §10.6 below).

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What separates the pattern from the adder is therefore not memory but the width of the coexistence window. The adder's bistability occurs at isolated inputs — one digit sum of the three available at m = 2 — because its control is discrete and has nothing in between. The pattern's occupies an interval of positive width, because both its control and its mismatch are continuous.

	Positional adder	**Pattern on a closed domain**
Preferred period	handed in as the base b	selected by growth rate or energy
Integer object	digit and carry	winding branch n
Mismatch	whole-number remainder, $0 \le r < b$	continuous strain δ
Coexistence of stable states	at isolated inputs only	over an interval of positive width
State change	deterministic carry rule	phase slip at a stability limit
Threshold	exact arithmetic	model- and parameter-dependent

9.5 What the adder is a model of

The earlier reading was that the adder is an exact model of a boundary and a poor model of a thing. This paper replaces it. The adder is the limit of pattern selection in which the period is handed in, the residue is a whole number, and the coexistence window has shrunk to isolated points. It is not a model of physics in general and does not claim that nature runs an adder. It says that when a system has a preferred continuous organization, an environment that admits only certain discrete organizations, a retained mismatch, and a threshold at which the mismatch forces a change of discrete organization, the arithmetic carry is what remains when the preference is fixed, the mismatch is quantized, and the window closes.

10. Closure, curvature, and the boundary that cannot talk to itself

10.1 The common hypothesis

Every theorem in [1]–[3] and in §§2–4 here uses four conditions.

- The positions are linearly ordered and the crossing state at any point is a single scalar carry.
- An interface between two regions is a run of unread positions characterized by its length g alone.
- No two crossing bonds are coupled to one another, because there is only ever one.
- Composition along an interface is serial and runs in one direction.

Under these, b^g controls everything cleanly because g is the only number an interface has. There is no branching, no second channel, no loop inside a gap, and no way for one part of a boundary to constrain another. The gap has no shape. Every physics-facing reading of the series has been made from inside this condition.

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10.2 What one dimension hides

In one dimension a region's boundary is two points, so the crossing state is at most m^2 and reads as a constant of the geometry. For a region R of a d-dimensional lattice under the same local rule, the boundary ∂R has of order L^{d-1} crossing bonds each carrying m states, and

$$\log F \sim |\partial R| \cdot \log m + \text{ledger}$$

— extensive in the boundary, not the volume. That is an area law, and F(S) is a bond dimension in the sense a transfer matrix or tensor network uses the term. The product over runs is the one-dimensional shadow of a product over boundary bonds. If the bonds were independent, severance would need $b^g \geq m$ per bond, so the depth threshold would be free of dimension while the cost scaled with area. They are not independent in $d \geq 2$: the bonds along a boundary are coupled by the same local rule, resetting one may conflict with resetting its neighbour, and neither the per-bond threshold nor the product law survives as stated (open problem 3).

10.3 No outside and no global frame

A finite lattice has edges, places where a rule stops applying and something has to be supplied from outside. An infinite plane has no edge but is non-compact: there is always further to go. A sphere is compact and has no boundary at all. In flat space translations commute, so a direction carried anywhere comes back unchanged and can serve as a reference. On a sphere a direction carried around a loop returns rotated by the enclosed curvature, $\Delta\theta = A/R^2$ for area A on a sphere of radius R. Marks persist; what fails is a global frame. A draft said nothing stays put under transport on a sphere; the correct statement is narrower.

A closed mesh of hexagons and pentagons with three edges at every vertex satisfies $V - E + F = 2$, $2E = 3V$ and $2E = 5P + 6H$, and these leave exactly $P = 12$ at any number of hexagons. More generally, with coordination numbers z_i, the disclination charges $q_i = 6 - z_i$ sum to $6\chi = 12$ on a sphere. Twelve is a floor on net charge, not a count of defects: large spherical crystals add neutral five–seven pairs, which gather into grain-boundary scars observed in colloidal crystals on droplets [16]. A draft said twelve defects; the correct statement is net charge twelve. Curvature itself is local and needs no embedding — at a vertex it is the angular deficit, 2π minus the angles meeting there, and the deficits sum to $2\pi\chi$.

10.4 A directed process on a closed surface must halt

A draft said a sphere has no halting points. That is true of the surface and false of anything moving across it. The carry has a direction, and on a sphere a direction field cannot be nowhere zero: by Poincaré–Hopf [19] the indices of a flow's zeros sum to $\chi = 2$ for every flow. A flow needs at least a place to start and a place to end, and by Reeb's theorem a closed surface that admits a function with only those two critical points is a sphere.

The line has exactly two halts, c_0 pinned at the bottom and the top carry discarded. It is the sphere's minimal flow opened to an outside. That is why §6 of [3] found the pinned origin to be a boundary condition and not a convention: a directed process on a closed world has to start somewhere, and the line makes it say where. The torus, with $\chi = 0$, admits a flow that never stops and a hexagonal mesh with no defects at all; it is the ring's world one dimension up. The corpus has studied the open case and the closed case with $\chi = 0$, and never the closed case with $\chi \neq 0$. Whether a reader's natural cuts on such a surface are forced to its critical points is open problem 5.

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10.5 Direction belongs to the reader

Lemma 1's normal form is associative in the block decomposition, and associativity is exactly the licence to evaluate a composition as a tree rather than a chain. That is the parallel-prefix property behind carry-lookahead addition [10]: all n carries computed in depth $\lceil \log_2 n \rceil$ rather than n − 1 sequential steps, at a cost of O(n log n) gates. The n − 1 of the ripple, and the n − 1 of [2] Corollary 4, are properties of a reader that walks. The machine is willing to be evaluated everywhere at once.

10.6 The ring

The ring is the one construction in the corpus that removes a clause of §10.1, and it carries the only residual: reading every position still leaves two classes, the double zero of one's-complement arithmetic ([3] §5.3). A draft asserted that the absence of a boundary implies a residual; closure preserves a residual that the dynamics produces but does not produce one, and a boundaryless system can have a unique consistent state. The residual here comes from the loop composite's fixed points. With a uniform word the loop map is the generator's n-th power; at b = 2, m = 2, t = 1 it is the identity on two states, and at m = 3 and m = 4 the loop maps (0,1,1), (1,1,2) and (0,1,1,1), (1,1,2,2), (2,2,2,3) are idempotents with two fixed points each. The sphere's twelve and the ring's two both arise from closure, but one comes from the topology and holds for any rule, and the other comes from the rule.

11. The algebra, briefly

Drafts of this work spent considerable effort on the carry-transfer monoid, the set of maps on C generated by $c \mapsto \lfloor (t + c)/b \rfloor$. It is reduced here to what bears on the rest.

It does not commute. At b = 2, m = 4 two single positions already disagree:

map	word	action on (0, 1, 2, 3)
$u(c) = \lfloor c/2 \rfloor$	t = 0	(0, 0, 1, 1)
$v(c) = \lfloor (c+1)/2 \rfloor$	t = 1	(0, 1, 1, 2)
$u \circ v$	v then u	(0, 0, 0, 1)
$v \circ u$	u then v	(0, 0, 1, 1)

In general the composite offset is $A + A'b^g$ one way and $A' + Ab^{g'}$ the other. The difference is visible to a reader, since the two composites send carry 2 to 0 and to 1. Order survives, but the composite of any history is named by a depth and an offset, so a reader of fixed size never has to hold the word. A reading of this work proposed finite state, non-commuting maps and reader-visible order dependence as the entry condition for a deeper system than the adder. The adder satisfies all three, so the condition separates nothing.

It is aperiodic. Every generator is nondecreasing in c, so every composite is, and the monoid lies inside the order-preserving maps of an m-chain, of which there are C(2m−1, m). At b = 2 the monoid has 3, 8, 16, 28 and 43 elements for m = 2 … 6, against 3, 10, 35, 126 and 462 order-preserving maps and 4, 27, 256, 3125 and 46656 maps in all. A finite monoid is aperiodic exactly when $x^n = x^{n+1}$ for every element, with n its size, and

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this holds in 20 of 20 configurations at b = 2…5, m = 2…6. The reason is general: a subgroup of a transformation monoid acts as bijections of an idempotent's image, the bijections here are order-preserving, and the only order-preserving bijection of a finite chain is the identity. In Krohn–Rhodes terms [11] the complexity is 0.

Three consequences follow. First, no group-valued conserved quantity can occur in this machine at any m, b, depth or gap structure; that search was closed by monotonicity before it began. Second, memory is not excluded. Krohn–Rhodes decomposes a finite semigroup into flip-flops, which reset and remember, and finite simple groups, which permute, and the flip-flop is aperiodic. The adder's bistable inputs of §9.4 are memory of exactly this kind, so the absence of a group layer and the absence of hysteresis are different facts. Third, the ring runs the same local rule, so it has the same aperiodic monoid, and its residual is a fixed-point count (§10.6), not a group.

Three instruments were tried and failed. Growth of the number of distinct maps with depth cannot distinguish anything, because a finite monoid is bounded by construction. The least number of parameters needed to name a composite is always 1, since one index can enumerate the monoid. Monoid size measures how narrow a family is, not its complexity; $\mathbb{Z}/5$ has five elements and complexity 1. Whether Krohn–Rhodes complexity is computable for an arbitrary finite semigroup is a long-standing open question, so it is used here only as a binary, decided by the aperiodicity test. A necessary condition for a group layer, and so for any group-valued conserved quantity, is that no single linear order on the state set makes every generator order-preserving. The adder fails it.

12. The specification and the overlay

The inversion that motivated this series — transformation first, matter as retained constraint, change as a projection — is not established by anything here. It is a specification, and a system instantiates it only by exhibiting five objects by name.

1. **The transformation family**: a state set and its admissible local maps, listed.
2. **The crossing state**: what must pass between two regions for them to stay coupled, and how much of it there is.
3. **The interface transfer family**: the maps an interface of given depth can realise on the crossing state.
4. **The retention or reset condition**: exactly when the interface can sever the dependence, and when it cannot.
5. **The observable quotient**: what a partial observer must retain, as a count, and how it composes.

A system that exhibits all five has an observer theory. One that exhibits fewer has an analogy. Applying the rule honestly empties most of the table below. A cell is filled only if the object is exhibited with a value, named if it exists in a literature but has not been computed in these terms, and empty otherwise.

System	Family	Crossing state	Transfers	Reset condition	Observer
m-operand base-b adder	filled	filled: m	filled: $\tau_{g,A}$	filled: $b^g \geq m$	filled: F(S), block law

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System	Family	Crossing state	Transfers	Reset condition	Observer
b-riffle shuffle	filled: same matrix	filled	filled	filled	empty
Swift–Hohenberg ring	filled	filled: winding n	named: phase slips	filled: Eckhaus limit, window measured	empty
Redundant signed digits	filled	filled: bounded	named	filled: depth 1	empty
Synchronizing automata	filled	filled	filled: semigroup	filled: reset word	empty
Tensor networks	named	named: bond dimension	named: gauge	empty	named: entanglement
Renormalization	named	named	named	empty	empty
Thermodynamics	empty	empty	empty	empty	empty
Quantum mechanics	empty	named, by counting only	empty	empty	named
Spacetime and locality	empty	empty	empty	empty	empty
A particle	empty	empty	empty	empty	empty

The Swift–Hohenberg ring is new to the table and is the first row outside the adder with four slots filled; its empty observer column is open problem 2. The synchronizing-automata row places the adder: its reset word has length $\lceil \log_b m \rceil$, logarithmic, against the quadratic $(m-1)^2$ of Černý's family [20]. Thermodynamics is empty and should stay so, because ln m here is Hartley information and nothing in the machine has a temperature. Quantum mechanics is empty except for bond-dimension counting, and the machine is irreversible by Theorem A‴ of [2]: a reader quotient that is reversible is one with no state. The particle row is the one that matters most, and it is empty: the only residual in the corpus is the ring's fixed-point count, which is not a persistent excitation, is not localized, and cannot interact with a second one.

13. Corrections

A killed claim is a constraint, so the ledger is part of the result. Thirty-five entries are recorded: eight against the earlier papers, twenty-three against drafts of this work, and four against readings of it. Several draft corrections were first raised by readings, and are credited to them in §15.

Against the earlier papers

1. [3]'s conditional product law is the all-gaps-wide case of the block law, and its open composition step is closed (§3).

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2. [3]'s capacity theorem reads $b^g \geq m$ as the gap realising a constant. It realises every constant; the forced set is all of C or empty (§3.1).
3. [3] describes a wide gap as forgetting the carry. The crossing state is reallocated to the gap word, and transmit + impose = m + 1 (§4).
4. [3] §8 reports the uniform coarsening fibre as an evenly spread loss. It is a change of units (§4.2, §6.1).
5. [3] §7 files base conversion as a curiosity. It is the division that generates every positional representation (§5.1).
6. [3] open problem 3 treats the ring's residual as unexplained. It is the fixed-point count of the loop composite, and the line shows the same bistability at isolated inputs (§9.4, §10.6).
7. The series presents the carry chain's base-free alphabet, b^{-j} contraction and base-invariant transfer family as its own. They are Holte's [5] (§6.1).
8. The series reads the adder as a model of a persistent thing, and later as a model of a boundary. It is a limit of pattern selection (§9.5).

Against drafts of this work

9. The lower block's state was identified with its interface. False, 96 of 111; repaired with the top quotient, 148 of 148 (§3.3).
10. Base change was called address-only, so that the machine has no depth. Base change rescales the grading (§6.1).
11. An interface's rank was said to decay. It is transferred, not depleted (§4.1).
12. Absence of a boundary was said to imply a residual. Closure preserves a residual; it does not produce one (§10.6).
13. F = 1, base-universality and equilibrium were stated as equivalent. They are equivalent on the pinned line only. This repeats correction 7 of [3], which was already on that paper's ledger when the error was made again.
14. $\Sigma \log_2 F$ was called an action and an entropy. It is a memory–time integral, bits held summed over steps; total erasure over a traversal does not depend on the order.
15. ln m was called the entropy of the carry. It is Hartley information; Shannon entropy equals it only under a uniform law, and the law is Eulerian (§4.4, §6.2).
16. An analog medium was said to supply resolution for free. It pays in restoring force, settling, tolerance, drift and energy (§5.4).
17. Pointing was called free and forgetting costly. Both cost when the gap word is charged; the difference is depth against selection (§4.3).
18. Growth in the number of distinct transfer maps was proposed as a test of depth. A finite monoid is bounded by construction (§11).
19. The least number of parameters naming a composite was proposed instead. It is always 1 (§11).
20. Krohn–Rhodes was summarized as two primitives. It is one reset component and every finite simple group dividing the semigroup (§11).
21. Monoid size was offered as a proxy for complexity. It measures restriction (§11).

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22. The ring's residual was predicted to carry a group. Running it: the ring has the line's aperiodic monoid (§10.6, §11).
23. Behaviour was said always to live in the remainder. The general form is repetition index plus local phase (§7.1).
24. Nothing was said to stay put under transport on a sphere. Marks persist; global frames do not (§10.3).
25. A sphere was said to have no halting points. Any flow on it has forced halts (§10.4).
26. Topology was said to force exactly twelve defects. It forces net charge twelve (§10.3).
27. The pattern's mismatch was written as N = qB + r. The mismatch is continuous; the branch is discrete (§9.1).
28. The hysteresis width was read as how long a count is kept. It is a range of the control parameter (§9.3).
29. The adder was said to have no dependence on history. It is bistable at isolated inputs; the difference from the pattern is window width (§9.4).
30. Citations: a "§13" of [3] was cited and does not exist, the passage being in [3] §10; the two Propositions 1 of [2] and [3] were conflated; Corollary 4 was attributed to [3] rather than [2].
31. Edits to a document were reported as made when no edit had been made. The rule adopted is to read back before reporting.

Against readings of this work

32. Finite state, non-commuting maps and reader-visible order dependence were proposed as the entry condition for a system deeper than the adder. The adder satisfies all three (§11).
33. Flat transport was said to have Krohn–Rhodes complexity zero. The translations of a torus form a group; the adder's zero comes from contraction, not flatness.
34. Mercury's resonance was written $3T_{orbit} = 2T_{spin}$. Its day is about 58.6 days and its year 88, so $2T_{orbit} = 3T_{spin}$ (§8.4).
35. The absence of hysteresis was equated with the absence of a group layer. The flip-flop, the canonical memory element, is aperiodic; memory and groups are different primitives (§11).

14. Open problems

Ordered by how much solving them would move, not by difficulty.

1. **The observer theory of a shuffle.** The carry chain is the descent chain of b-riffle shuffles. Define an access set on a shuffle, the quotient of histories it induces, and whether the count factorizes across a separation. The dynamics is identical and fully characterized, so any failure to transfer is a fact about the cut theory. It is the cheapest test of generality available.
2. **The observer theory of a pattern-forming ring.** The Swift–Hohenberg ring has the carry with a coexistence window of positive width. What must a reader of part of the ring hold, and does its state count depend on which branch the ring occupies? It is the natural next system: classical dynamics, a carry that is present, and a window that is measurable.
3. **Two coupled bonds.** The smallest violation of §10.1: an interface with two crossing bonds sharing a cell. Does per-bond separation compose, and if not, what replaces the product?

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4. **The factorization curve.** The fraction of cuts that factorize against $g / \log_b m$ across many (b, m). Collapse with a knee near one half would connect guaranteed severance to Eulerian concentration (§6.2).
5. **Reader cuts on a closed surface.** On a surface with $\chi \neq 0$ every directed flow has forced critical points. Are a reader's natural cuts forced to them? The torus against the sphere, with the same rule and mesh construction, is the controlled experiment (§10.4).
6. **The upper bound of the splitting lemma as an induction on n** (§3.3).
7. **A group layer small enough to compute.** A finite family meeting the necessary condition of §11, whose observer theory can still be computed exactly.
8. **Growing m.** Every result fixes m. With m = m(n) the threshold grows with the system and the boundary between coupling and separation moves with scale.
9. **Retained paths.** The number of distinct carry walks reaching a fixed target grows exponentially — 1, 3, 30, 236, 1966, 15961 at n = 4 and 1, 11, 111, 1709, 23682, 336284 at n = 5 [3] — while the endpoint retains none of it. Which machines retain a path in a bounded state is not addressed in the corpus and cannot be addressed in this machine.

Closed here: open problems 1 (composition) and 2 (the ramp) of [3]. Explained but not closed: open problem 3 of [3] (the ring's residual) and the one-way implication of [3] correction 12.

15. Method

A default mistaken for a signature. The same failure has now produced five results. An operand count of 2 was read as a binary carry, a radix of 2 as base-free, $c_0 = 0$ as a convention, a tape loading as the absence of a coordinate, and a dimension of 1 as the absence of geometry. In each case the defective claim was true where it was made and could not be found false by checking it there. Each was found by moving a parameter nobody had noticed was a parameter.

The programme passed its own saturation depth. By §2.3 the reach saturates at $\lceil \log_b m \rceil$, and past that depth more positions add nothing. The same arithmetic applies to the method. More resolution on the adder stopped adding information several papers ago, and drafts of this work kept adding it — monoids, aperiodicity, Krohn–Rhodes, fixed-point counts. Those results are true and occupy one section. To learn more, change the system, not the magnification.

Agreement is not evidence of scope. Independent readers evaluate a claim where it was made, so reproducing it tests the argument and not its reach. Three readings of [3] reproduced $\lceil \log_b m \rceil$ without varying b. In this work readings correctly raised several of the draft corrections — Hartley against Shannon, the no-boundary overreach, the plane's missing edge, marks against frames, net charge against defect count, the type of the pattern's mismatch, and the hysteresis window as a range. They also introduced the four errors recorded against readings. Agreement carries errors forward as readily as results; only independent parameterization supports a claim.

A literature check is the cheapest parameter move. Holte's 1997 paper contains what several papers of this series rederived. The check is two-sided: see what the higher layer exposes, then see what it hides. Holte supplied the chain and nothing about partial access, and that absence is where this series' contribution lies.

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Verification kills extensions; it did not choose a question. The results here rest on executed checks with no failures, and several attractive claims died against them. None of the reframings came from a sweep. They came from a two-block question in place of a k-run one, a corrected sign on a decay that was a transfer, one lookup, a remark that transformation happens in every direction at once, a remark about how a base is actually changed, and a remark about gravity.

Two triggers. Suspect a result that is too clean for its question, and suspect independent readings that agree — both are signs of an unmoved default.

Read back, and read the ledger. One draft reported edits as made when none had been; another repeated an error already on [3]'s ledger. Before reporting a state, read it back. Before extending a result, read the corrections.

16. Reproduction

Every number in this paper was produced by an executed run during its preparation, and every count attributed to [1]–[3] was checked against the text of those papers.

Result	Outcome	Scope
Splitting lemma	455 of 455	27 (b, m, n) configurations, $n \le 8$
Upper bound at every gap	1,151 of 1,151	all two-block splits, wide or narrow
Direction of run-product failure	431 over, 0 under	same sweep
Block law	1,349 of 1,349 (919 / 405 / 25 blocks)	same sweep
Setter, pointer, separation	105 of 105 each	$b = 2…6$, $m = 2…8$, $g = 1…3$
Reach closed form	1,848 of 1,848	$b = 2…12$, $m = 1…24$, $L = 0…6$
Ramp saturation at $\lceil \log_b m \rceil$	418 of 418; 480 of 480 by depth	$b = 2…12$, $m = 2…39$
Top-truncation quotient	722 of 722	$b = 2…6$, $m = 2…12$, $n = 2…5$
Interface identification	96 of 111 (false); 148 of 148 repaired	$b \le 5$, $n \le 5$
Aperiodicity of the transfer monoid	20 of 20	$b = 2…5$, $m = 2…6$
π, hexadecimal to decimal	exact; BBP 22 of 22 digits	first 22 hex digits
Swift–Hohenberg ring	switching values of §9.2–9.3	$N = 256$, $L = 16\pi$, $r = 0.3$, $dt = 0.05$, seeded

The Swift–Hohenberg runs use a semi-implicit spectral step with the linear operator implicit and the cubic term explicit, noise of amplitude 10^{-6} per step from a fixed seed, a ramp of 120,000 steps in each direction, and cell counts taken as upward zero crossings every 200 steps. They are deterministic under that seed.

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